Building Vibration Measurement Analysis

Turbine-Shutdown Addendum:

Direct Confirmation of Heating-Plant Coupling

Evidence:	Direct turbine-operating and shutdown comparison
Primary comparisons:	Five paired floor and source-path locations
Physics floor pairs:	Titan, Herzog, and PHYS 244
Report date:	July 29, 2026

Supplement to the Building Vibration Measurement Analysis

1 Direct shutdown evidence in the Physics building

The turbine-operating spectra are shown on the left and the corresponding shutdown spectra on the right. Every paired panel uses the same frequency range and logarithmic amplitude scale. Paired floor spectra were available for Titan, Herzog, and PHYS 244. PHYS 111 and PHYS 245 are omitted because no measurement was obtained for these locations during the unscheduled shutdown. However it should be noted from the original report that the spectrum of PHYS 245 should be comparable to PHYS 244 as before due to proximity.

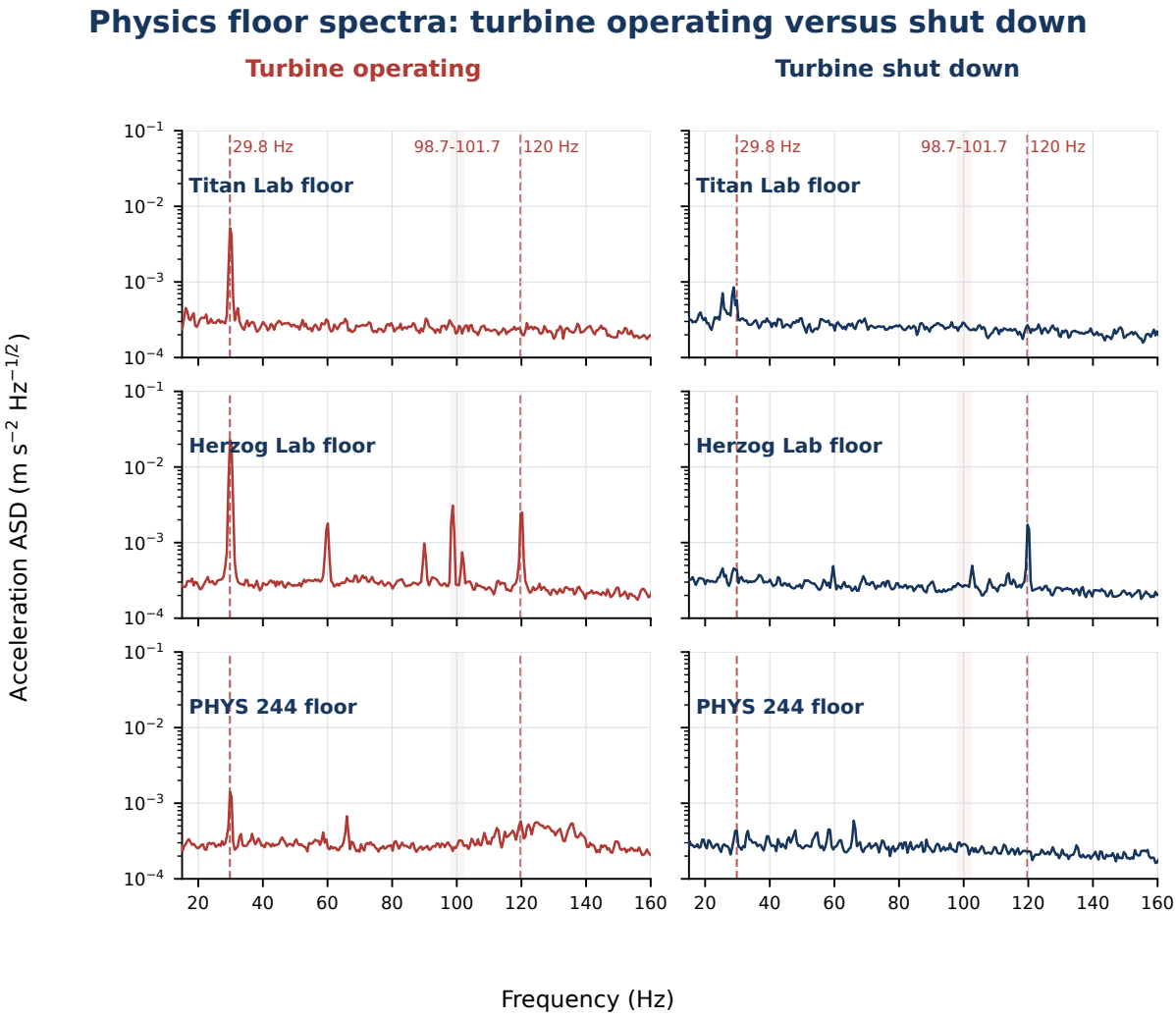


Figure 1: Paired Physics floor spectra on common axes. Vertical red guides mark the 29.8 Hz line, the 98.7–101.7 Hz group, and the approximately 120 Hz line identified in the operating-state report. The 29.8 Hz feature collapses in all three shutdown measurements. The 120 Hz peak that remains in Herzog during shutdown is examined separately below.

At 29.8 Hz the shutdown/on ASD changes are -19.0 dB at Titan, -34.4 dB at Herzog, and -10.3 dB at PHYS 244. The same line disappears from the source-path measurements below. This is direct confirmation that the previously observed 29.8 Hz building line was generated by the turbine.

2 The source and propagation path

The pipe and sidewalk comparison makes the source attribution decisive. The large operating-state peaks at 29.8 Hz, 98.7–101.7 Hz, and about 120 Hz are present at the plant and along the route toward Physics, then are absent when the turbine is shut down.

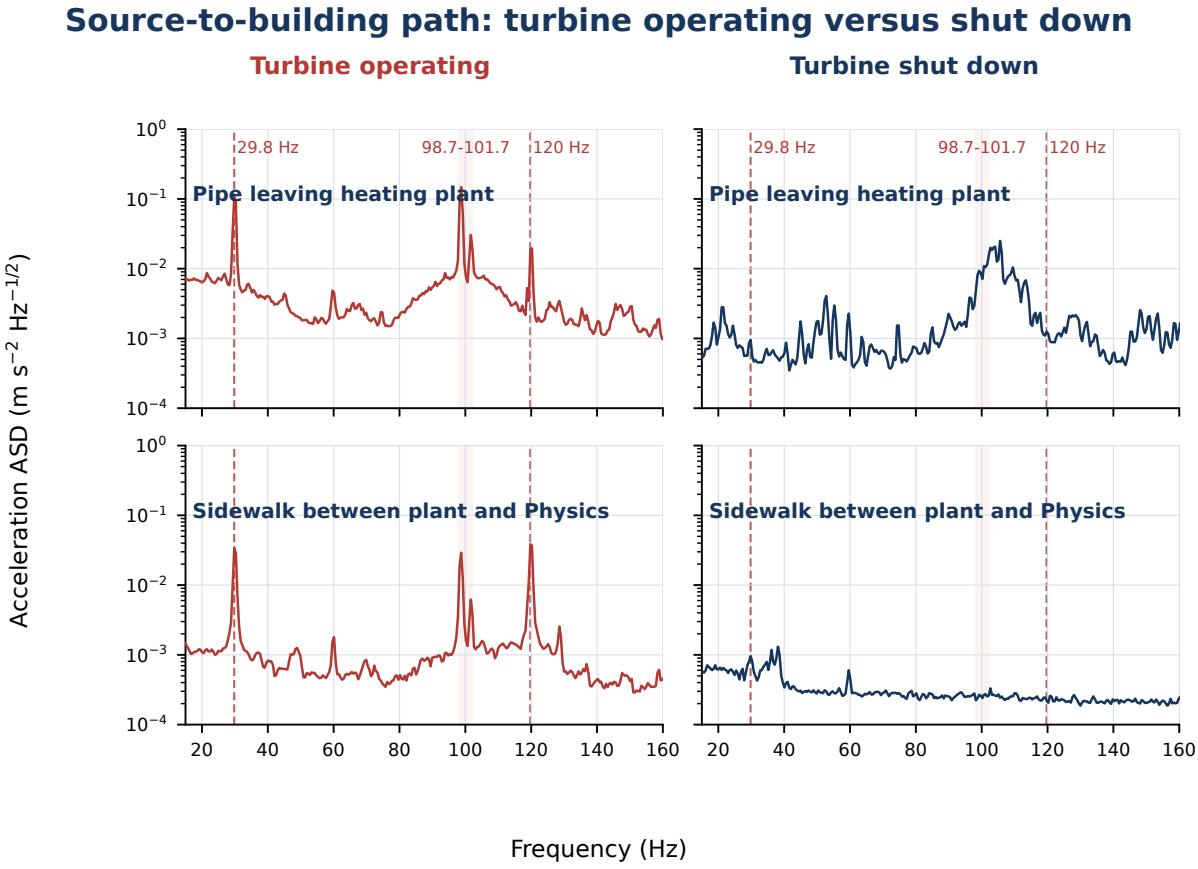


Figure 2: Source-path spectra with operating and shutdown states placed side by side on identical axes. The operating-state line pattern is carried from the plant pipe to the sidewalk. The disappearance of the external 29.8 Hz, 98.7–101.7 Hz, and 120 Hz components during shutdown identifies them as heating-plant features.

Table 1: Primary reductions. Full-band RMS uses the same 0.5–250 Hz definition as the main report. The last column is the shutdown/on change at 29.8 Hz; negative values are reductions.

Location	Full-band RMS (10^{-3} m s^{-2})		Reduction (%)		29.8 Hz
	On	Off	Full	20–50 Hz	change (dB)
Titan Lab floor	6.02	3.79	37.1	61.6	-19.0
Herzog Lab floor	22.3	4.22	81.1	91.9	-34.4
PHYS 244 floor	4.52	3.75	17.2	20.5	-10.3
Pipe leaving heating plant	176	49.7	71.7	94.9	-41.1
Sidewalk between plant and Physics	59.8	5.08	91.5	90.1	-31.0

3 Turbine lines and the residual Herzog 120 Hz peak

The shutdown test separates turbine signatures from unrelated peaks. A line is assigned to the turbine where it is strong at the plant and sidewalk during operation and disappears or falls sharply when the turbine stops.

Table 2: Shutdown/on ASD change in dB at the comparison frequencies. Negative values are reductions. The persistent Herzog value near 120 Hz is distinguished below from the external heating-plant component.

Line (Hz)	Pipe	Sidewalk	Titan	Herzog	PHYS 244
29.8	-41.1	-31.0	-19.0	-34.4	-10.3
98.7	-25.2	-41.7	-0.3	-20.9	2.3
101.7	-8.5	-27.8	1.1	-9.2	-3.6
119.7	-23.6	-44.9	1.2	-2.9	-8.1

The 29.8 Hz line is definitively turbine-generated: it falls by 41.1 dB at the pipe and 31.0 dB at the sidewalk, collapsing down to the noise floor of the measurment instrument in every paired spectra.

The 98.7–101.7 Hz group is also a turbine source signature. It likewise falls below the noise floor along the source path and at locations where they were prominent inside Physics.

The external approximately 120 Hz component is turbine-related. The 119.7 Hz amplitude falls by 23.6 dB at the pipe and effectively below the noise floor on the sidewalk adjacent to the Herzog lab, so the heating-plant contribution is no longer present. However the peak persist in the Herzog floor albeit at reduced intensity. Because 120 Hz is the common second harmonic of 60 Hz mains and can be generated by motors and compressors, the operating ultra-cold freezer is the likely local source.

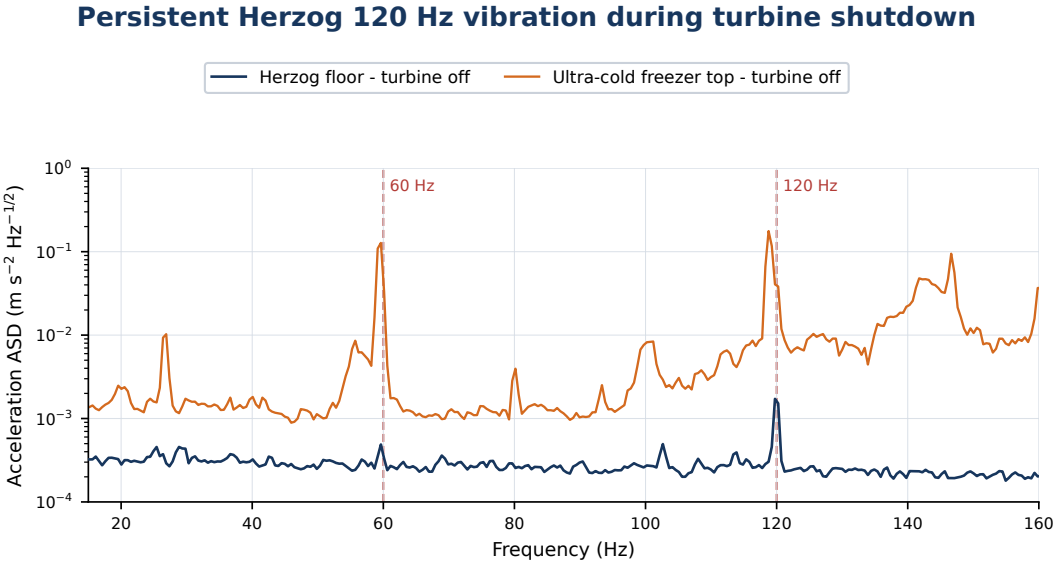


Figure 3: Turbine-off spectra measured on the Herzog floor and on top of the operating ultra-cold freezer. The external heating-plant-associated 120 Hz component is absent in the source-path shutdown measurements, but the persistent Herzog floor peak aligns closely with the freezer spectrum. This makes the freezer the most likely source of the remaining laboratory vibration near 120 Hz; a measurement with the freezer powered down is required for definitive confirmation.

4 Conclusion

The shutdown showed more than removal of isolated lines. Full-band RMS acceleration falls by 71.7% at the plant pipe and 91.5% on the sidewalk, while the paired room floors fall by 37.1% at Titan, 81.1% at Herzog, and 17.2% at PHYS 244. These reductions show that the turbine produced the clearly identified narrow lines and a substantial part of the elevated integrated vibration background measured along the plant-to-Physics path.

The 29.8 Hz line and 98.7–101.7 Hz source group originate with the heating-plant turbine. The external approximately 120 Hz component is also turbine-related, but the 120 Hz line remaining inside Herzog during shutdown has a separate local origin. Its close frequency alignment with the operating ultra-cold freezer makes that equipment the leading candidate, although this attribution remains provisional until a freezer-off measurement is made. However, as that frequency doesn't persist through the optical tables and the freezer could always be relocated, it is likely not be a question worth perusing.